

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Develop a Dynamic College Tech Fest Registration Webpage using JavaScript and DOM

2. AIM

To develop a dynamic event registration webpage using HTML, CSS and JavaScript DOM manipulation that validates user input and dynamically displays the submitted registration information.

3. OBJECTIVE

The objectives of this experiment are:

- Create an attractive College Tech Fest registration interface.
 - Use HTML input and select elements to collect participant information.
 - Use JavaScript functions to perform registration and reset operations.
 - Access and modify webpage elements using `document.getElementById()`.
 - Validate the required fields before processing registration.
 - Generate a unique registration ID using `Math.random()` and `Math.floor()`.
 - Display the participant name, email, team, event and current date/time dynamically.
 - Clear the form values and hide the result section using the reset operation.
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4. THEORY

JavaScript is a scripting language that adds dynamic behaviour and interactivity to webpages. In this experiment, JavaScript is used to respond to button clicks, read values entered by the participant, validate the information and update the webpage dynamically.

The Document Object Model (DOM) represents the HTML document as a collection of objects. JavaScript can access and modify these objects using methods such as `document.getElementById()`.

The program uses DOM manipulation to retrieve input values, change the visibility of the

result section and insert registration information dynamically using innerHTML. JavaScript features used in the program include let variables, functions, conditional statements, Math.random(), Math.floor(), the Date object, toLocaleString(), string concatenation and dynamic HTML generation.

CSS is used to create the visual appearance of the registration interface. The webpage contains a gradient background, centered registration container, styled input fields, selection boxes, buttons, focus effects, hover effects, borders, shadows and a fade animation.

5. ALGORITHM / PROCEDURE

1. Create the HTML document and define the character encoding, viewport and webpage title.
 2. Design the College Tech Fest registration interface using heading, input fields, select elements and buttons.
 3. Apply CSS styling to the webpage, form controls, buttons and result section.
 4. Define the register() function for processing the registration.
 5. Retrieve the name, email, team and event values using document.getElementById().
 6. Remove unnecessary spaces from the name and email values using trim().
 7. Check whether any required field is empty.
 8. If any field is empty, display an alert asking the participant to complete all fields and stop the registration process.
 9. Generate a registration ID using the CTF- prefix and a random four-digit number.
 10. Create a Date object to obtain the current local date and time.
 11. Make the result section visible.
 12. Use innerHTML to display the generated registration ID and participant details.
 13. Display a successful registration message.
 14. Define the resetForm() function to clear the input fields and restore the default selections.
 15. Hide the result section after resetting the form.
 16. Open the webpage in a browser and verify the registration, validation and reset operations.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>College Tech Fest Registration</title>
```

```
<style>

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, sans-serif;
}

body{
  background:linear-gradient(to bottom,#00111a,#003344,#00111a);
  color:white;
  display:flex;
  justify-content:center;
  align-items:center;
  min-height:100vh;
}

.container{
  width:450px;
  background:rgba(15,25,30,0.95);
  padding:25px;
  border-radius:15px;
  box-shadow:0 0 20px cyan;
  border:2px solid #00bcd4;
}

h1{
  text-align:center;
  color:#00e5ff;
  text-shadow:0 0 10px cyan;
  margin-bottom:5px;
}

p{
  text-align:center;
  color:#ddd;
  margin-bottom:20px;
}

input,select{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:none;
  border-radius:6px;
  background:#263238;
  color:white;
  font-size:15px;
}

input:focus,
select:focus{
  outline:none;
  box-shadow:0 0 10px cyan;
}

button{
```

```
width:48%;
padding:12px;
border:none;
border-radius:6px;
margin-top:15px;
cursor:pointer;
font-size:15px;
}

.register{
background:#0097a7;
color:white;
}

.register:hover{
background:#00bcd4;
}

.reset{
background:#444;
color:white;
}

.reset:hover{
background:#666;
}

#result{
display:none;
margin-top:20px;
padding:15px;
border-radius:10px;
background:#101820;
border:1px solid cyan;
animation:fade 0.5s;
}

@keyframes fade{
from{
opacity:0;
transform:scale(0.9);
}

to{
opacity:1;
transform:scale(1);
}
}

.success{
color:#00ff99;
font-size:20px;
text-align:center;
}

</style>
</head>

<body>
```

```
<div class="container">

<h1>COLLEGE TECH FEST</h1>

<p>Annual Technical Event Registration</p>

<input type="text" id="name" placeholder="Enter Your Name">

<input type="email" id="email" placeholder="Enter Email">

<select id="team">

<option value="">Choose Your Team</option>
<option>Code Masters</option>
<option>Tech Titans</option>
<option>Digital Innovators</option>
<option>Future Coders</option>

</select>

<select id="event">

<option value="">Choose Event</option>
<option>Web Development</option>
<option>Code Challenge</option>
<option>Project Expo</option>
<option>Quiz Competition</option>

</select>

<div style="display:flex;justify-content:space-between;">

<button class="register" onclick="register()">Register</button>

<button class="reset" onclick="resetForm()">Reset</button>

</div>

<div id="result"></div>

</div>

<script>

function register(){

let name=document.getElementById("name").value.trim();

let email=document.getElementById("email").value.trim();

let team=document.getElementById("team").value;

let event=document.getElementById("event").value;

if(name=="" || email=="" || team=="" || event==""){

    alert("Please complete all the fields.");
```

```

        return;
    }

    let id="CTF-"+Math.floor(1000+Math.random()*9000);

    let now=new Date();

    document.getElementById("result").style.display="block";

    document.getElementById("result").innerHTML=

    "<h2 class='success'>Registration Completed!</h2><br>"+

    "<b>Registration ID:</b> "+id+"<br><br>"+

    "<b>Name:</b> "+name+"<br>"+

    "<b>Email:</b> "+email+"<br>"+

    "<b>Team:</b> "+team+"<br>"+

    "<b>Event:</b> "+event+"<br>"+

    "<b>Date:</b> "+now.toLocaleString()+"<br><br>"+

    "<span style='color:#00e5ff;'>Welcome, "+name+
    "! Get ready to participate in the College Tech Fest.</span>";

    }

    function resetForm(){

    document.getElementById("name").value="";

    document.getElementById("email").value="";

    document.getElementById("team").selectedIndex=0;

    document.getElementById("event").selectedIndex=0;

    document.getElementById("result").style.display="none";

    }

</script>

</body>

</html>

```

7. CODE EXPLANATION

Code	Explanation
HTML document structure	Defines the page with the DOCTYPE, html, head and body elements, along with meta tags and the page title.
CSS styling	Uses selectors, a dark linear gradient, flexbox, colours, borders, shadows, focus and hover states, and a fade animation.
Form controls	Uses text and email input fields and two select elements for choosing a team and event.
JavaScript functions	Defines register() for registration processing and resetForm() for clearing and resetting the form.
DOM manipulation	Uses document.getElementById() to access the form controls and the result element.
Input validation	Checks whether name, email, team or event is empty. If any field is empty, an alert is displayed and the function returns.
Random registration ID	Uses Math.random() and Math.floor() to generate a four-digit number with the ST- prefix.
Date and time	Creates a Date object and uses toLocaleString() to display the current local date and time.
Dynamic result	Changes the result element to visible and uses innerHTML to insert the registration ID and submitted details.
Reset operation	Clears the input values, resets both select elements to their first option and hides the result element.

8. TEST CASES AND EXECUTION DATA

The following test cases are based directly on the validation, registration and reset logic present in the source program.

Test Case	Input / Action	Expected Result	Status
TC-01	Leave one or more fields empty and click Register.	Alert "Please fill all fields." is displayed and registration is stopped.	Pass
TC-02	Enter name, email, team and event and click Register.	Registration result is displayed with generated ID, entered details and current date/time.	Pass

Test Case	Input / Action	Expected Result	Status
TC-03	After registration, click Reset.	Input fields are cleared, selections are reset and the result is hidden.	Pass
TC-04	Change the selected team and event and register with valid inputs.	The selected team and event appear in the dynamic result.	Pass

9. OUTPUT

Output 1: Stranger Things Event Registration after successful registration

The screenshot displays a registration form for 'COLLEGE TECH FEST' with the subtitle 'Annual Technical Event Registration'. The form fields are filled with the following information:

- Name: Hansikha
- Email: Hansikha@gmail.com
- Team: Future Coders
- Event: Web Development

Below the form fields are two buttons: 'Register' (highlighted in red) and 'Reset' (disabled). Below the buttons, a confirmation box displays the following details:

- Registration Completed!**
- Registration ID: CTF-7884
- Name: Hansikha
- Email: Hansikha@gmail.com
- Team: Future Coders
- Event: Web Development
- Date: 7/9/2026, 4:21:35 pm
- Welcome, Hansikha! Get ready to participate in the College Tech Fest.

The output shows the completed registration form with the entered details, selected team and event, generated registration ID, current date/time and dynamic welcome message.

10. RESULT

Thus, a dynamic College Tech Fest Registration webpage was successfully developed using HTML, CSS and JavaScript. The application validates the required inputs, generates a registration ID, displays the submitted information dynamically using DOM manipulation, and provides a reset operation for clearing the form.